

Heart Disease Prediction Using Machine Learning Techniques

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Abstract—Heart disease is one of the leading causes of death worldwide and continues to be a major public health concern. Early diagnosis and treatment can significantly reduce mortality rates and improve patient outcomes. This project presents a machine learning based heart disease prediction system using different classification algorithms including Random Forest, Logistic Regression, Decision Tree, and K-Nearest Neighbors (KNN). The dataset was preprocessed and analyzed using data mining techniques such as feature scaling, visualization, and classification analysis. The performance of different models was evaluated using accuracy, precision, recall, F1-score, and confusion matrix analysis. Experimental results demonstrated that the Random Forest algorithm achieved the highest accuracy of 85%, making it the best performing model for heart disease prediction. The proposed system can assist healthcare professionals in predicting heart disease efficiently and support early medical decision making.

Index Terms—Heart Disease Prediction, Machine Learning, Data Mining, Random Forest, Classification Algorithms, Healthcare Analytics

I. INTRODUCTION

Heart disease is one of the most dangerous and life-threatening diseases affecting millions of people globally. According to global health studies, cardiovascular diseases are responsible for a significant percentage of deaths every year. Factors such as unhealthy lifestyle, high cholesterol, smoking, obesity, stress, and lack of physical activity contribute to the increasing number of heart disease cases.

Traditional methods of heart disease diagnosis may require extensive medical examinations and expert analysis. With the advancement of machine learning and data mining techniques, intelligent healthcare systems can now analyze patient data and provide predictions with high accuracy. Machine learning algorithms can identify hidden patterns within medical datasets and assist doctors in making better clinical decisions.

This project focuses on developing a heart disease prediction system using machine learning algorithms. The system analyzes various medical parameters such as age, blood pressure, cholesterol level, chest pain type, fasting blood sugar, and heart rate to determine whether a patient is likely to suffer from heart disease.

II. OBJECTIVES

The major objectives of this project are:

- To study and analyze heart disease datasets using data mining techniques.

- To preprocess and visualize healthcare data for better understanding.
- To implement multiple machine learning classification algorithms.
- To compare the performance of different machine learning models.
- To predict the presence of heart disease with improved accuracy.
- To develop a simple GUI based prediction system for user interaction.

III. LITERATURE REVIEW

Machine learning techniques have become increasingly important in healthcare applications. Several researchers have applied classification algorithms for disease prediction and diagnosis. Logistic Regression is commonly used for binary classification problems due to its simplicity and interpretability. Decision Tree algorithms are popular because they are easy to understand and visualize.

Random Forest is an ensemble learning technique that combines multiple decision trees to improve prediction performance and reduce overfitting. K-Nearest Neighbors (KNN) is another classification technique that predicts output based on similarity between data points.

Previous studies have shown that machine learning models can effectively predict heart disease using patient medical attributes. Among various algorithms, Random Forest often provides better performance due to its robustness and ability to handle complex datasets.

IV. DATASET DESCRIPTION

The dataset used in this project contains medical information related to heart disease patients. It includes several clinical and physiological attributes that are useful for disease prediction.

Important features included in the dataset are:

- Age
- Sex
- Chest Pain Type
- Resting Blood Pressure
- Cholesterol Level
- Fasting Blood Sugar
- Resting ECG Results
- Maximum Heart Rate Achieved

- Exercise Induced Angina
- Oldpeak
- Slope
- Number of Major Vessels
- Thalassemia
- Target Variable

The target variable contains:

- 0 = No Heart Disease
- 1 = Presence of Heart Disease

V. METHODOLOGY

A. Data Preprocessing

Data preprocessing is an important step in machine learning because raw datasets may contain inconsistencies or unnecessary information. The following preprocessing steps were performed:

- Checking for missing values
- Data cleaning and formatting
- Feature scaling using StandardScaler
- Splitting the dataset into training and testing sets

The dataset was divided into:

- 80% training data
- 20% testing data

B. Machine Learning Algorithms

The following machine learning algorithms were implemented and compared:

- Random Forest
- Logistic Regression
- Decision Tree
- K-Nearest Neighbors (KNN)

C. Model Evaluation Metrics

The models were evaluated using the following performance metrics:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

VI. SYSTEM ARCHITECTURE

The overall workflow of the proposed system is shown below.

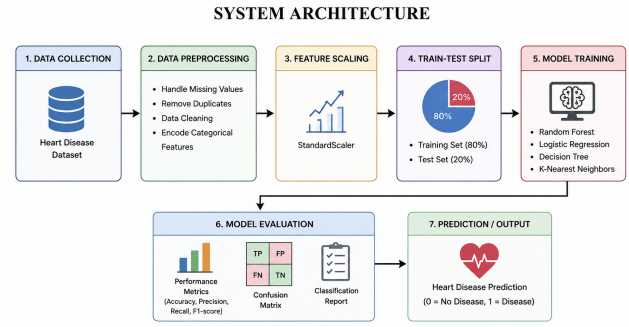


Fig. 1. System Architecture of Heart Disease Prediction System

The system begins with dataset collection and preprocessing. The processed data is then provided to machine learning algorithms for model training and prediction. Finally, the system evaluates model performance and predicts whether a patient has heart disease.

VII. IMPLEMENTATION

The project was implemented using Python programming language and several machine learning libraries.

Libraries used in this project include:

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Tkinter

The application consists of:

- Data preprocessing module
- Machine learning model module
- GUI based prediction system

VIII. DATA VISUALIZATION

A. Correlation Matrix

The correlation matrix heatmap helps identify relationships between different medical features and target variables. Strong positive or negative correlations indicate important features for prediction.

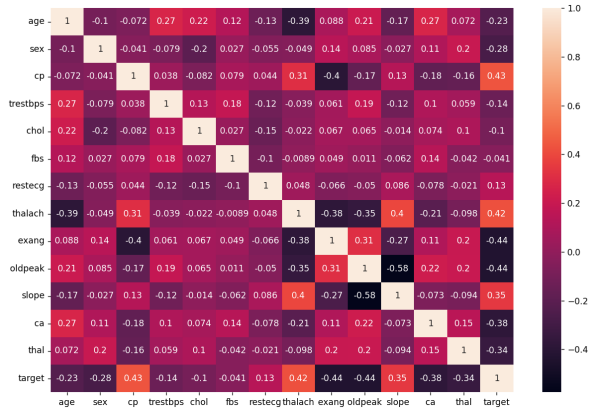


Fig. 2. Correlation Matrix Heatmap

B. Target Variable Distribution

The target distribution graph shows the number of patients with and without heart disease in the dataset.

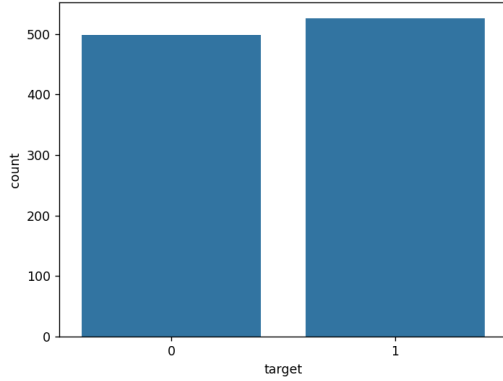


Fig. 3. Target Variable Distribution

IX. RESULTS AND DISCUSSION

Different machine learning algorithms were trained and tested using the heart disease dataset. Their performance was compared based on accuracy.

TABLE I
ACCURACY COMPARISON OF ALGORITHMS

Algorithm	Accuracy
Random Forest	85%
Logistic Regression	77%
Decision Tree	74%
KNN	74%

The Random Forest classifier achieved the highest prediction accuracy among all algorithms. This is because Random Forest combines multiple decision trees and reduces overfitting, resulting in better generalization performance.

A. Confusion Matrix

The confusion matrix of the Random Forest model is shown below.

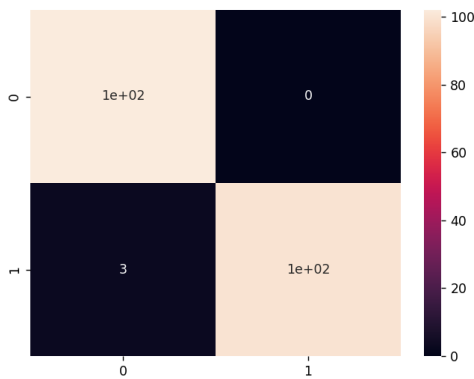


Fig. 4. Confusion Matrix for Random Forest

The confusion matrix shows that the model correctly classified most positive and negative heart disease cases.

B. Classification Report

TABLE II
CLASSIFICATION REPORT FOR RANDOM FOREST

Class	Precision	Recall	F1-score
0	0.90	0.81	0.85
1	0.81	0.90	0.85

The Random Forest algorithm achieved balanced precision and recall values, indicating reliable prediction performance for heart disease classification.

X. GUI BASED PREDICTION SYSTEM

A simple Graphical User Interface (GUI) was developed using Tkinter. The GUI allows users to enter patient medical parameters and receive prediction results instantly. The interface improves usability and demonstrates practical application of machine learning in healthcare systems.

The GUI predicts:

- Heart Disease Detected
- No Heart Disease

XI. ADVANTAGES OF THE PROPOSED SYSTEM

- Early prediction of heart disease
- Reduced diagnosis time
- Improved healthcare decision support
- Better prediction accuracy
- User-friendly GUI interface
- Efficient medical data analysis

XII. LIMITATIONS

Although the system performs effectively, some limitations still exist:

- Limited dataset size
- Prediction accuracy depends on dataset quality
- Cannot completely replace professional medical diagnosis
- Results may vary with different datasets

XIII. FUTURE SCOPE

Future improvements can further enhance the system capabilities:

- Integration of deep learning techniques
- Development of web-based applications
- Real-time patient monitoring systems
- Integration with hospital databases
- Prediction of multiple diseases
- Mobile healthcare applications

XIV. CONCLUSION

This project successfully developed a machine learning based heart disease prediction system using multiple classification algorithms. Various data preprocessing and visualization techniques were applied to analyze the dataset effectively. Among all implemented algorithms, Random Forest achieved the highest accuracy of 85%, making it the best performing model for prediction.

The developed GUI application demonstrates how machine learning can be applied in healthcare systems to assist doctors and patients in disease prediction. The project highlights the importance of intelligent healthcare systems and the role of machine learning in improving medical diagnosis and decision making.

REFERENCES

- [1] UCI Machine Learning Repository, "Heart Disease Dataset."
- [2] Scikit-learn Documentation. Available: <https://scikit-learn.org>
- [3] Ian H. Witten, Eibe Frank, Mark A. Hall, "Data Mining: Practical Machine Learning Tools and Techniques."
- [4] J. Han and M. Kamber, "Data Mining Concepts and Techniques," Morgan Kaufmann.
- [5] Python Software Foundation. Available: <https://www.python.org>
- [6] T. Mitchell, "Machine Learning," McGraw-Hill Education.
- [7] F. Chollet, "Deep Learning with Python," Manning Publications.